MIL-R-42010A
19 February 1988
SUPERSEDING
MIL-R-42010 (DM)
10 June 1986

MILITARY SPECIFICATION

RETRACTOR, NERVE ROOT (HOLSHER)

This specification is approved for use for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of nerve root retractors of the Holsher design.

1.2 Classification. Retractors shall be of the following sizes as indicated below (see 6.2):

Size	
1	1/4 inch tip
2	1/2 inch tip

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

F 1089 - Standard Test Method for Corrosion of Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Retractors shall be manufactured from Class 4 martensitic corrosion-resistant steel. As an alternate, the retractor or retractor handle may be manufactured from Class 3 austenitic corrosion-resistant steel. The chemical composition of the steel shall be in accordance with ASTM F899 for the class and type used. Composition shall be tested as specified in 4.5.1.

3.2 Hardness. After proper stress relieving, retractors manufactured from martensitic corrosion-resistant steel shall be hardened to a Rockwell hardness of HRC 38-45. Austenitic corrosion-resistant steel shall have a Rockwell hardness of HRB 80-95. Hardness shall be tested as specified in 4.5.2.

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3.3 Style, design and dimensions. The retractors shall be 8.5 inches plus or minus 0.25 inches in length. The retractor shall be of the style and design shown in Figure 1, and shall have the dimensions shown in Figure 1. As an alternate, the shank of the retractor may be round or octagonal. The handle shall be round stock 3/16 plus or minus 1/32 inches in diameter formed in an open loop. The handle shall be 3-1/2 inches plus or minus 1/8 inches in length. The distal 1-3/4 plus or minus 1/8 inches of the retractor shall be formed at an angle of 33^0 plus or minus 2^0 relative to the centerline of the shank. The concave side of the distal retractor tip shall have a minimum depth of 1/16 inch for the size 1 instrument and 3/32 inch for the size 2 instrument.

3.4 Markings. The shank of the retractor shall bear the manufacturer's name or registered trademark and the word "stainless." Instruments of foreign manufacture shall bear the name of the country of manufacture.

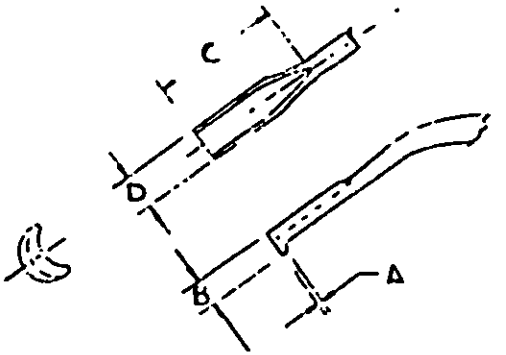
3.5 Passivation. The retractor shall be properly passivated.

3.6 Finish. The finish shall be entirely mirror, satin or matte. The matte finish shall exhibit a uniform non-directional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure. Mirror or satin finishes shall be polished uniformly.

3.7 Corrosion. Retractors shall show no signs of corrosion when tested as specified in 4.5.4.1. In addition, retractors manufactured from martensitic corrosion-resistant steel or retractors which have a section manufactured from martensitic corrosion-resistant steel shall show no plating of copper when tested as specified in 4.5.4.2. Plating at the periphery of drops or dulling of polished surfaces shall be disregarded.

3.8 Workmanship. The retractor shall be free from sharp edges, pits, crevices, and burrs, grind and tool marks, or any defect which affects the serviceability or utility of the retractor.

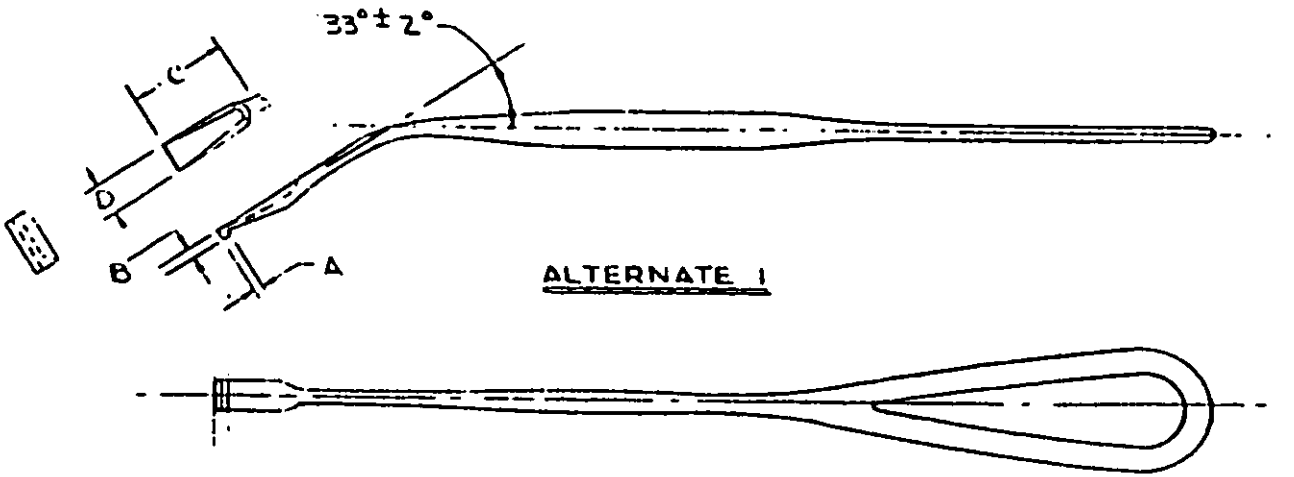
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ALTERNATE 2

(HANDLE SAME AS ALTERNATE 1.)

	SIZE	A ± 0.25	B ± 0.25	C ± 0.25	D ± 0.8
ALTERNATE 1	1	1.27	2.36	20.00	6.4
	2	1.83	2.67	22.00	12.7
ALTERNATE 2	1	1.10	4.50	25.40	7.0
	2	1.10	7.00	28.57	13.0



ALTERNATE 1

NOTE:

ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

FIGURE 1. RETRACTOR, NERVE ROOT

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Governnment to acceptance of defective material.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.2.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2.2 Certificate of quality. Certificate of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of composition testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one retractor.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one retractor. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Composition	3.1	4.5.1	1/	2/
Hardness	3.2	4.5.2	S-2	1.0
Dimensions	3.3	---	S-2	2.5
Corrosion	3.7	4.5.3	S-2	1.0

1/ A sufficient amount of raw material shall be tested.

2/ The lot shall be rejected if test requirements are not met.

4.4 Examination.

4.4.1 Examination of retractors. The retractors shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any retractor in the sample containing one or more defects shall be rejected and if the number of defective retractors in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Retractor not of the specified design for appropriate type, stock number and item identification.
102	Retractors not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
103	Finish of retractor not uniform and not as specified.
104	Retractor has grind or tool parts.
105	Identification markings not present, not complete or not legible.
Minor	
201	Retractor not free from dirt, polishing material or foreign matter.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Composition. Raw material shall be tested for composition in accordance with Federal Test Method 111 or 112 of Federal Test Method Standard Number 151 (see 4.2.2).

4.5.2 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5.3 Corrosion tests. Retractors shall be tested for resistance to corrosion as specified in 3.8 and 4.5.3.1 and 4.5.3.2.

4.5.3.1 Boil test. The retractor shall be tested in accordance with the resistance to boil test of ASTM Standard F1089.

4.5.3.2 Copper sulfate test. The retractor shall be tested in accordance with the resistance to copper sulfate test of ASTM standard F1089.

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4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Retractors shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit Package. Each retractor shall be packaged in accordance with MIL-P-116, Method IC-2.

5.1.1.4 Intermediate package. Twelve unit packages containing retractors of one size only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, Class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for retractors shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages containing retractors of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Twelve intermediate packages containing retractors of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for retractors shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall confrom to type IV or type V.

5.2.5.1 Level A and B. Retractors packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Retractors packed as specified in 5.2.3 shall be unitized as specified 5.2.5.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

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6. NOTES

6.1 Intended use. The nerve root retractor shall be used during neurosurgical and orthopedic spinal surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

6.3 This specification is intended to cover the following National Stock Numbers:

National Stock No.	Item Identification	Specification Designation
6515-01-164-8855	RETRACTOR, NERVE ROOT, HOLSHER, 8-1/2 Inches Long, 1/2 Inch Tip	2
6515-01-164-8856	RETRACTOR, NERVE ROOT, HOLSHER, 8-1/2 Inches Long, 1/4 Inch Tip	1

6.4 Subject term (key word) listing.

Holsher retractor
Nerve retractor
Root retractor
Retractor

6.5 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Retractors of matte finish shall have a reduced reflective surface that is more glare resistance than satin finish.

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3767

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-42010A	RETRACTOR, NERVE ROOT (HOLSHER)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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